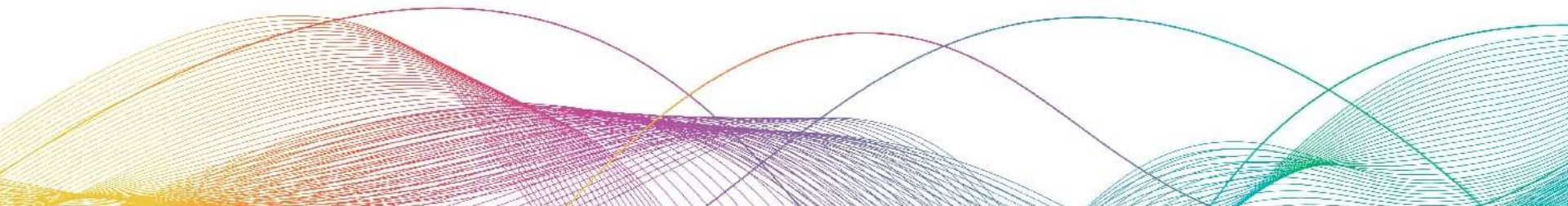


06 Social



Social

Implement a web application with basic social functions, including user accounts, friend management, instant messaging, and post sharing.

Technology Stack:

Layer	Technology	Description
Frontend	HTML + CSS + Vanilla JavaScript	No framework, easy to learn and understand
Backend	Flask (Python)	Lightweight web framework
Real-time Communication	WebSocket (SocketIO)	Implement instant messaging
Database	JSON files	Lightweight data storage
Responsive Layout	CSS Media Queries	Adapt to mobile/desktop

Phase 1: Registration/Login + Personal Settings

Implement user registration, login, and profile settings.

Page	File	Description
Login Page	login.html	User login
Registration Page	register.html	User registration
Settings Page	settings.html	Profile management

The image displays two side-by-side mockups of a user interface for a social media application, specifically focusing on the registration and login phases.

Left Mockup (Login Page):

- Header: A blue circular icon with a white face, followed by the text "欢迎回来" (Welcome Back) and "登录你的社交媒体" (Log in to your social media).
- Form Fields:
 - 用户名 (Username):** A text input field with a placeholder "请输入用户名" (Please enter username).
 - 密码 (Password):** A text input field with a placeholder "请输入密码" (Please enter password).
- Buttons:
 - A large black button labeled "登录" (Login).
 - A link below the button: "还没有账户? 立即注册" (Don't have an account? Register now).

Right Mockup (Registration Page):

- Header: A blue circular icon with a white face, followed by the text "创建账号" (Create Account) and "加入我们, 开始社交" (Join us, start social).
- Form Fields:
 - 用户名 (Username):** A text input field with a placeholder "4-16个字符, 字母/数字/下划线" (4-16 characters, letters/numbers/underscores).
 - 密码 (Password):** A text input field with a placeholder "至少6位密码" (At least 6 characters password).
 - 确认密码 (Confirm Password):** A text input field with a placeholder "请再次输入密码" (Please enter password again).
- Buttons:
 - A large black button labeled "注册" (Register).
 - A link below the button: "已有账号? 返回登录" (Already have an account? Return to login).

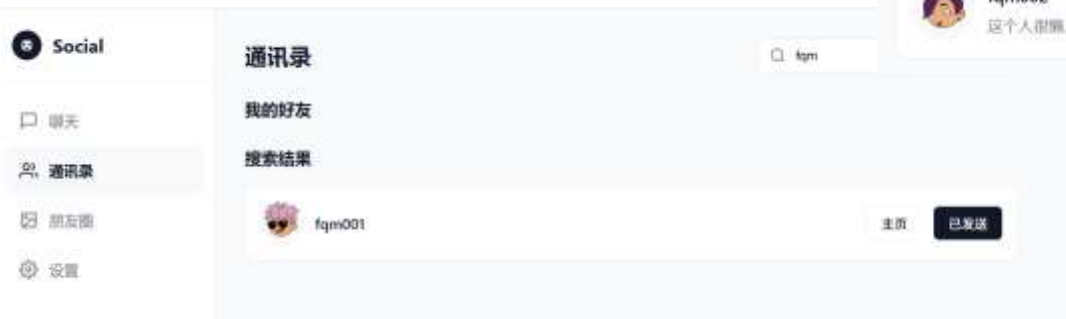
Phase 2: Contacts + Friend Management

Implement contacts page, user search, send friend requests, manage friend requests.

| Page | File | Description |

|-----|-----|-----|

| Contacts Page | contacts.html | Friend list + search + friend requests |



Phase 3: Instant Messaging

Implement real-time chat feature, including message sending, receiving, and conversation list.

| Page | File | Description |

|-----|-----|-----|

| Messages Page | index.html#chats | Chat conversation list |

| Chat Detail Page | chat.html | Chat detail |

聊天

Q 搜索聊天...



fqm003

你好

10:27



fqm002

Hi

10:26



fqm003

你好

10:27

Phase 4: Moments/Posts

Implement post publishing, browsing
and interactions (likes, comments).

Page	File	Description
Discover Page	friendcircle.html	Post list
Profile Page	profile.html	User's posts

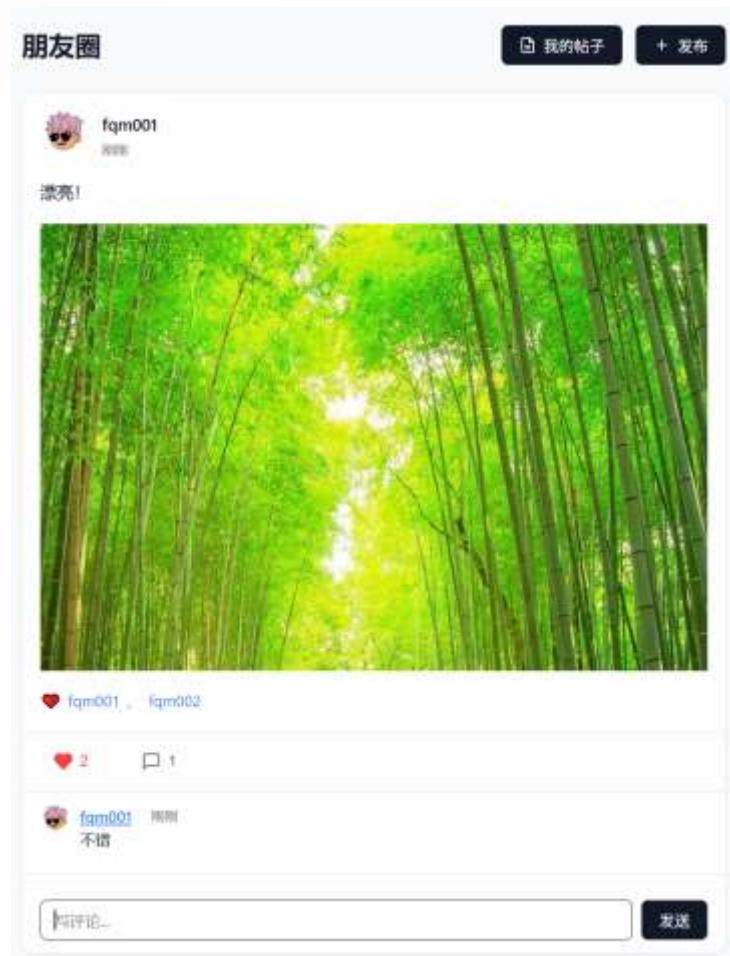
发布朋友圈

分享新鲜事...

添加图片

取消

发布



Phase 5: Admin Backend

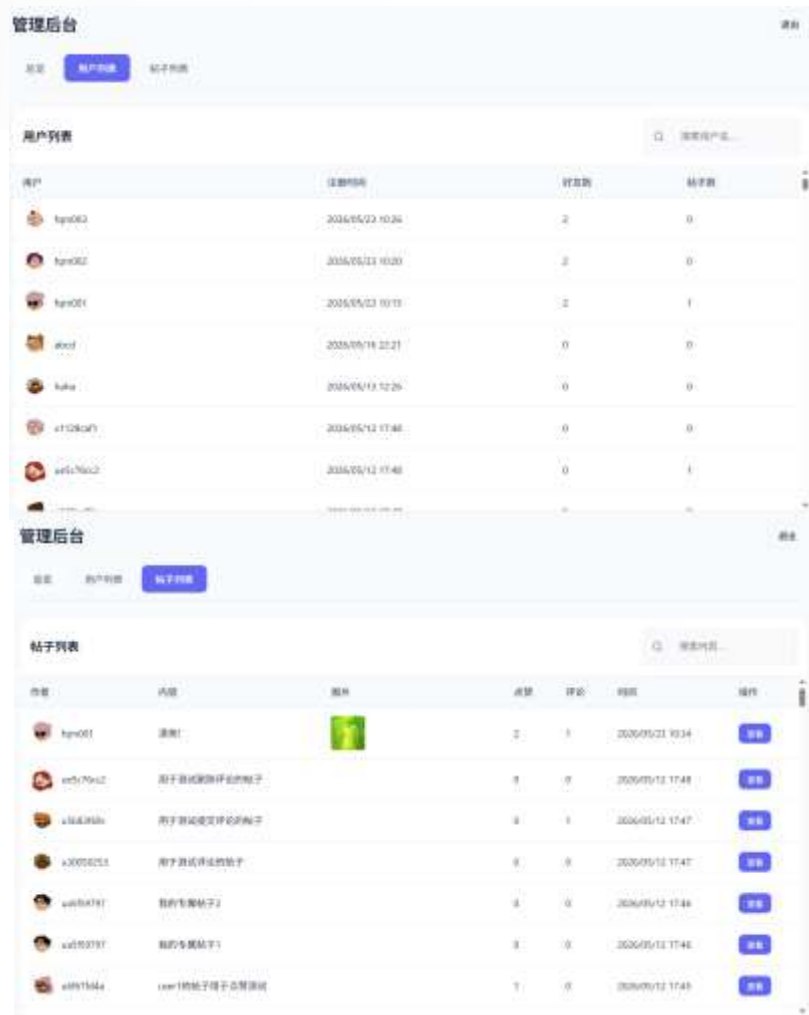
Implement admin backend for managing users and posts.

| Page | File | Description |

|-----|-----|-----|

| Admin Login Page | admin.html | Admin login |

| Admin Backend | admin.html | Statistics + user management + post management |



The screenshot shows the '管理后台' (Admin Backend) interface with two tables: '用户列表' (User List) and '帖子列表' (Post List).

用户列表 (User List):

用户	注册时间	好友数	帖子数
hgm002	2024/05/22 10:26	2	0
hgm002	2024/05/22 10:20	2	0
hgm001	2024/05/22 10:13	2	1
aliof	2024/05/18 22:21	0	0
haha	2024/05/12 12:26	0	0
u1128ca9	2024/05/12 11:48	0	0
u01c79a2	2024/05/12 11:48	0	1

帖子列表 (Post List):

作者	内容	图片	点赞	评论	时间	操作
hgm001	漂亮!		2	1	2024/05/21 10:14	删除
u01c79a2	用于测试删除评论的帖子		0	0	2024/05/12 11:48	删除
u1128ca9	用于测试删除评论的帖子		0	0	2024/05/12 11:47	删除
u0070213	用于测试评论的帖子		0	0	2024/05/12 11:47	删除
u01c79a2	我的专属帖子2		0	0	2024/05/12 11:48	删除
u01c79a2	我的专属帖子1		0	0	2024/05/12 11:48	删除
u01c79a2	user1的帖子得点赞测试		1	0	2024/05/12 11:48	删除

Automated Testing

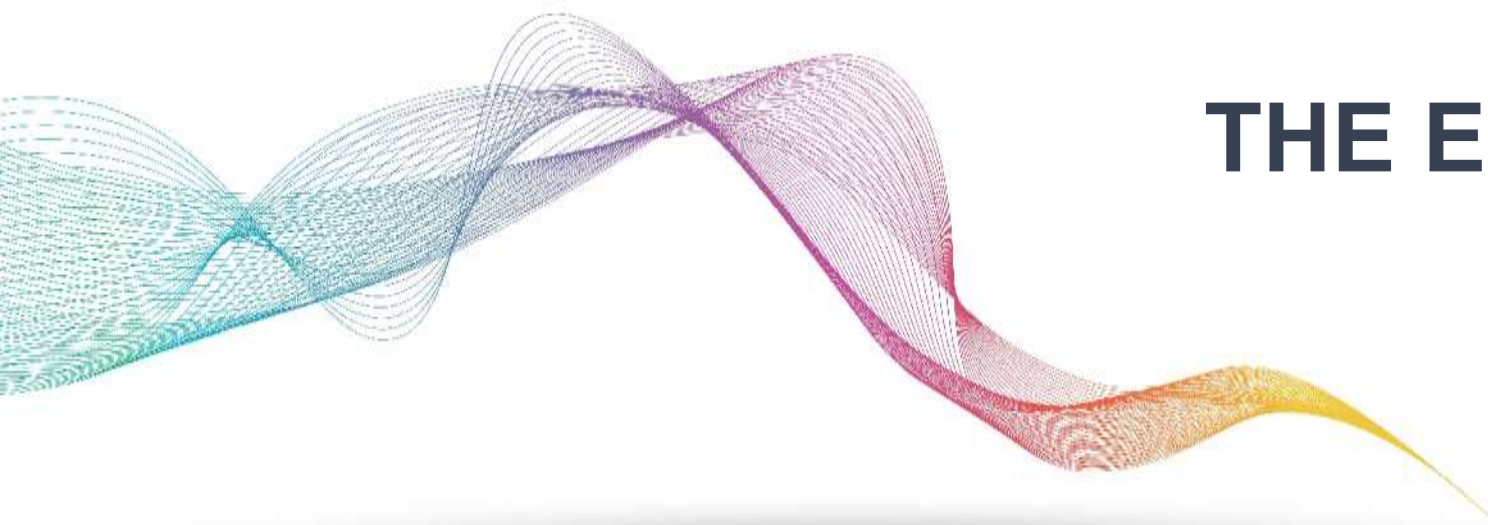
Steps:

1. Write a test checklist using AI.
2. Use AI to install testing environment dependencies (When first used).
3. Create test file structure using AI.
4. Conduct API testing
5. Conduct UI testing
6. Run all tests (some tests require manual completion)
7. View test results

```
EN.md  TEST_CHECKLIST.md *
tests > TEST_CHECKLIST.md > # Social 测试清单
# Social 测试清单
## 前提 -> 登录注册 + 个人信息
### API 测试
| 5 | 用户注册-用户名已存在 | 'test_register_duplicate_username' | ■ |
| 6 | 用户登录成功 | 'test_login_success' | ■ |
| 7 | 用户登录-密码错误 | 'test_login_wrong_password' | ■ |
| 8 | 用户登录-用户不存在 | 'test_login_user_not_found' | ■ |
| 9 | 退出登录 | 'test_logout' | ■ |
| 10 | 获取当前用户信息 | 'test_get_me' | ■ |
| 11 | 获取指定用户信息 | 'test_get_user' | ■ |
| 12 | 更新个人资料-简介 | 'test_update_bio' | ■ |
| 13 | 更新个人资料-头像 | 'test_update_avatar' | ■ |
| 14 | 修改密码-成功 | 'test_change_password_success' | ■ |
| 15 | 修改密码-当前密码错误 | 'test_change_password_wrong' | ■ |
| 16 | 修改密码-新密码过短 | 'test_change_password_short' | ■ |
| 17 | 修改密码-两次输入不一致 | 'test_change_password_mismatch' | ■ |

### UI 测试
| 序号 | 测试用例 | 测试函数 | 状态 |
|-----|-----|-----|-----|
| 1 | 登录页面加载 | 'test_login_page_loads' | ■ |
| 2 | 登录页面元素验证 | 'test_login_page_elements' | ■ |
| 3 | 注册页面加载 | 'test_register_page_loads' | ■ |
| 4 | 注册页面元素验证 | 'test_register_page_elements' | ■ |
| 5 | 登录流程 | 'test_login_flow' | ■ |
| 6 | 注册流程 | 'test_register_flow' | ■ |
| 7 | 设置页面加载 | 'test_settings_page_loads' | ■ |
| 8 | 修改头像 | 'test_change_avatar' | ■ |
| 9 | 修改个人资料 | 'test_change_bio' | ■ |
| 10 | 修改密码 | 'test_change_password' | ■ |
| 11 | 退出登录 | 'test_logout' | ■ |
```

```
tests
├── __pycache__
│   ├── __init__.cpython-314.pyc
│   └── conftest.cpython-314-pytest-9.0.3...
├── api
│   ├── __pycache__
│   │   ├── __init__.py
│   │   ├── test_admin.py
│   │   ├── test_auth.py
│   │   ├── test_friends.py
│   │   ├── test_messages.py
│   │   ├── test_posts.py
│   │   └── test_users.py
├── ui
│   ├── __pycache__
│   │   ├── __init__.py
│   │   ├── test_admin_pages.py
│   │   ├── test_auth_pages.py
│   │   ├── test_chat.py
│   │   ├── test_contacts.py
│   │   ├── test_posts.py
│   │   ├── test_red_dot.py
│   │   ├── __init__.py
│   │   ├── conftest.py
│   │   ├── run_all_tests.py
│   └── TEST_CHECKLIST.md
```

THE END